Self-Supervised Divide-and-Conquer Generative Adversarial Network for Classification of Hyperspectral Images

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***Abstract*— Generative adversarial network (GAN) has been rapidly developed because of its powerful generating ability. However, imbalanced class distribution of hyperspectral images (HSIs) easily causes mode collapse in GAN. Moreover, limited training samples in HSIs restrict the generating ability of GAN. These issues may further deteriorate the classification performance of the discriminator. To conquer these issues, a novel self-supervised divide-and-conquer (SDC)-GAN is proposed for HSI classification. In SDC-GAN, a pretext cluster task with an encoder–decoder architecture is designed by leveraging abundant unlabeled samples. By transferring the learned cluster representation from the cluster task, limited labeled samples are divided effectively in the downstream classification. According to the division of clustering, SDC-GAN constructs a generic and several specific branches for both the generator and discriminator. The generator generates all- and specific-class samples by using the generic and specific branches separately and combines them adaptively. It can weaken the generation preference for the classes with large sample sizes and alleviate the mode collapse problem. Meanwhile, the classification ability of the discriminator is improved by integrating the judgment of specific branches into the generic branch. Experimental results show that SDC-GAN achieves competitive results for HSI classification compared with several state-of-the-art methods.**

***Index Terms*— Divide and conquer, generative adversarial networks (GANs), hyperspectral image (HSI) classification, self-supervised learning.**

I. Introduction

THE rapid development of remote sensing imaging technology makes it more convenient to acquire abundant and large-range remote sensing images. Hyperspectral (HS) sensors are capable of imaging at wavelengths ranging from visible to infrared spectrum [1]. The acquired HSI is a three-dimensional (3-D) data cube, in which a 2-D image can be obtained from each band and each pixel contains hundreds of spectral bands. Thus, a wealth of spatial and spectral information exists in HSIs simultaneously. In the past few decades, HSI processing technology has been successfully used in many real-world scenarios, for instance, military [2], precise agriculture [3], geological prospecting [4], and environmental monitoring [5], [6].

HSI classification has attracted much attention of various scholars from remote sensing communities. The HSI classification task is initially solved by using classic machine learning algorithms, such as decision trees [7], k-nearest neighbor [8], and support vector machines (SVMs) [9]. However, these algorithms heavily depend on expert knowledge and hand-crafted features, which is difficult to directly transfer in complex scenarios.

Since the 2010s, deep learning-based methods have shown extreme potential in natural image classification. Subsequently, some of these methods have been applied for the classification of HSIs. Different from the traditional classification methods, deep learning-based methods perform feature extraction and classifier training at the same time, and the hand-crafted features are no longer needed anymore. Chen *et al.* [10] proposed a stacked autoencoder (SAE) method for solving the classification problem of HSIs. It first uses principal component analysis (PCA) to reduce the dimensionality of high-dimensional HSI data, then trains a feature extractor to obtain high-quality features, and finally classifies the samples by a logistic regression classifier. Later, various improved versions were proposed based on SAE, such as sparse SAE [11], generative adversarial convolutional SAE [12], and so on. In 2015, a deep belief network (DBN) [13] was constructed by using the layer-by-layer learning manner of the restricted Boltzmann machines for HSI classification. The improved DBN methods include diversified DBN [14] and hierarchical cascade DBN [15]. Both two types of deep learning-based methods use the fully connected (FC) layer, resulting in too many trainable parameters, which is a severe challenge for HSI data with small a sample size.

Different from SAE and DBN, convolutional neural network (CNN) [16] has the characteristics of local connection and

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weight sharing, which not only considers spatial information but also decreases the number of trainable parameters. Subsequently, a battery of CNN-based approaches has emerged for HSI processing immediately [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30]. According to different kinds of extracted features, these approaches can be mainly grouped into three categories: spectral feature-based, spatial feature-based methods, and joint spectral–spatial feature-based approaches [17]. Hu *et al.* [18] introduced a 1-D CNN by acting on each original spectral signature directly. In [19], PCA was first exploited to reduce the dimensionality of HSIs, and 2-D CNN was performed to learn nonlinear, discriminative and invariant spatial features for HSI classification. Moreover, a 3-D CNN framework was proposed to learn joint spectral–spatial features of HSIs. Yu *et al.* [20] developed a new dense CNN framework for HSI classification. It combines dense networks with feedback attention modules for spatial and spectral feature extraction and classification. Then, deeper CNNs [21], [22] have become a mainstream trend for the extraction of high-order semantic features and further improvement of classification performance. However, deeper CNNs easily overfit with scarce training samples of HSIs. Li *et al.* [23] modified a deep pixel-pair feature (PPF)-based classification approach to mitigate the overfitting problem. In addition, some semisupervised CNNs [24], [25], [26] come forward one after another by utilizing the amounts of unlabeled data.

In the last decade, GAN has received widespread attention because of its competitive generation capability. Original GAN [31] contains a generator that generates the images by learning the distribution of real ones and a discriminator that attempts to classify the real and generated images. The original discriminator only distinguishes between true and fake images. Thus, it cannot be directly applied to the classification task. Odena *et al.* [32] designed an auxiliary classifier GAN (ACGAN), which could deal with the multiclassification task. In [33], deep convolutional GAN (DCGAN) was constructed to combine the advantages of CNN and GAN, which is widely used in later studies.

Numerous GAN-based HSI classification models have sprung up in recent years. The classification task of HSIs is aided by the generation of high-quality extended samples. Zhu *et al.* [34] designed a 1-D GAN and a 3-D GAN model, in which spectral and joint spectral–spatial features are extracted to classify the HSIs separately. In [35], a multiclass spatial–spectral GAN (MSGAN) was proposed, it designed two different generators to generate 1-D spectrum and 2-D spatial images, respectively, and the discriminator extracts joint spectral–spatial features and outputs the classification probabilities. Later, Wang *et al.* [36] proposed an adaptive dropblock-enhanced GAN (AD-GAN), which changes the two-branch output of the discriminator in ACGAN into a single branch and designs an adaptive dropblock strategy to improve the generalization ability. These GAN-based algorithms only use limited labeled samples to generate new samples, which restricts the generation ability of GAN to a certain extent.

In addition to supervised GAN-based methods, there are some semisupervised GAN methods that have been proposed for HS image (HSI) classification. HS-GAN [37] was proposed for HSI classification. In HS-GAN, 1-D GAN was constructed and trained by utilizing the unlabeled samples. Then, the discriminator is transformed into a classifier and finetuned with limited labeled samples. In [38], a semisupervised variational GAN is designed, which consists of the generator, discriminator, and classifier. It builds a collaborative relationship between the generator and classifier to generate discriminative samples for classification. Zhong *et al.* [39] proposed a two-step spectral–spatial GAN method, which integrates a semisupervised GAN and a conditional random field (CRF) for HSI classification. Liang *et al.* [40] proposed a semisupervised SSAT-GAN method by combining the classification loss of labeled samples with the reconstruction loss of unlabeled samples. These improved GAN-based methods use the same network to generate samples for all the categories at one time, which does not consider the differences in the difficulty of generating samples with different categories. In HSIs, imbalanced class distribution is serious, and the majority class has dozens or even hundreds of times more samples than the minority one. In this case, GANs prefer to generate the classes with large sample sizes in HSIs, which may easily cause the mode collapse.

In this article, a novel self-supervised divide-and-conquer GAN (SDC-GAN) model is proposed for solving the classification problem of HSIs. In SDC-GAN, a pretext cluster task is designed to leverage the amounts of unlabeled data and assist the downstream classification task. In the downstream task, the divide-and-conquer architecture is designed in the generator and discriminator. Specifically, an encoder–decoder network is designed in the cluster task, where more informative features are extracted for clustering in the encoder by the construction of the decoder. Then, labeled samples are divided by the learned cluster network from the pretext task. According to the clustering results, the generator and discriminator in SDC-GAN are constructed by designing a generic and several specific branches. In the generator, the generic branch and the specific branches are designed to generate all the samples and class-specific samples, and then, high-quality samples can be obtained by fusing the results of multiple branches. In the discriminator, the generic branch and the specific branches are used to extract the universal and more discriminative features, respectively. Furthermore, a new fused loss function is defined to optimize the generator and discriminator by combining multiple branches. Ultimately, the well-trained discriminator of SDC-GAN is utilized to classify the test HSI data.

The main contributions of this article can be summarized as follows.

1) Self-supervised learning is introduced in GAN for HSI classification for the first time. It achieves the precise division of limited labeled samples with the assistance of abundant unlabeled samples and provides the potential for the generator and discriminator to improve the generation and classification ability.
2) SDC-GAN divides the samples into multiple branches, makes the number of samples smaller, and the class distribution is relatively balanced in the specific branches, which can mitigate the problem of imbalanced class

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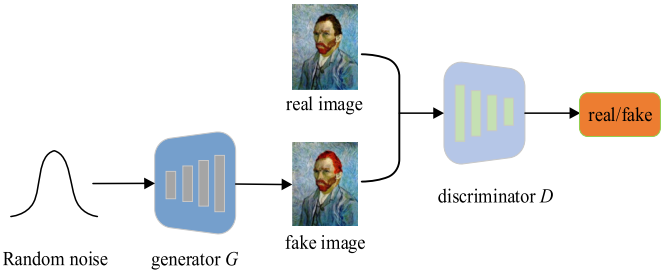


Fig. 1. Original GAN.

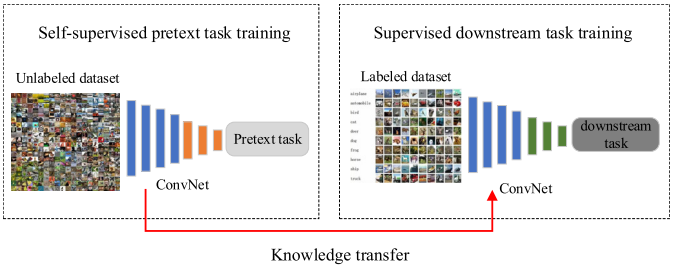


Fig. 2. Classic framework of self-supervised learning.

distribution. Meanwhile, it prompts the generator to improve the quality of generated samples by fusing the results from different branches adaptively.

3) SDC-GAN uses the 3-D input and the operation of cascading 1-D and 2-D convolution to extract the spatial–spectral features of HSIs. In addition, a multi-branch discriminator with multiple specific branches and a generic branch is designed, so that the hard samples with similar spectral features are reclassified by specific branches. In this case, the network is forced to mine the discriminative features of these hard samples and improve the performance of classifying these hard samples.

The remaining of this article consists of four sections. Among them, Section II introduces the background of GAN and self-supervised learning. The architecture of the SDC-GAN algorithm is described in Section III in detail. Subsequently, the experimental results of SDC-GAN and the comparison methods are analyzed in Section IV. Finally, some conclusion and future work are presented in Section V.

II. Background

A. Generative Adversarial Network

Based on game theory, GAN was first proposed by Sun *et al.* [27] as a new generative model in 2014. As illustrated in Fig. 1, GAN has two players: a generator G and a discriminator D. The input of G is the noise vector z sampled from Gaussian or Uniform distribution. The goal of G is to make the distribution of the generated data constantly approximate that of the real data x and its output $G(z)$ has the same shape as the real data. The inputs of D are x and $G(z)$, and the corresponding outputs are $D(x)$ and $D(G(z))$. The output of D denotes the probability that the current input comes from the real data distribution. The purpose of D is to distinguish the generated data and the real data as accurately as possible. Instead, G wants D to make mistakes and gives a high probability to $G(z)$.

The loss function of GAN is defined as follows:

$$\max_D \min_G V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log(D(x))\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log(1-D(G(z)))\right] \quad (1)$$

where p_{data} and p_z are the distributions of x and z. $\mathbb{E}(\cdot)$ indicates the empirical estimation. The global optimality of GAN is the Nash equilibrium of the mini-max game, which is achieved in an alternate training way. Specifically, given G, D is optimized by maximizing $V(D,G)$, then G is optimized by minimizing $V(D,G)$ with fixed D. Both G and D are improved gradually through adversarial training. After finishing the training, D is hard to discriminate the fake data from G and the real data x, which means that the quality of the fake data is good enough. In this case, the global optimal solution $p_g = p_{\text{data}}$ is obtained, where p_g is the distribution of generated data from G.

B. Self-Supervised Learning

Self-supervised learning belongs to unsupervised learning, which is proposed to extract general low-level features from massive unlabeled samples. It can avoid the enormous cost of collecting and annotating the data. To learn common and informative features from unlabeled samples, a classical approach is to design some appropriate pretext tasks for the networks to solve. By optimizing the objective function of the pretext tasks, the networks can learn general features by the guidance of the pseudo labels. The pretext tasks often have two characteristics: the first one is that the representative features of the data can be captured by solving the pretext task, and the other one is that the pseudo labels can be obtained automatically. The commonly used pretext tasks are image colorization [41], image inpainting [42], image jigsaw puzzle [43], and so on.

Fig. 2 shows the classic framework of self-supervised learning. Based on the attributes of the data, the pseudo labels are generated for a predefined pretext task automatically. During the training of pretext task, the network is devised to extract the representative features. After finishing the training of the pretext task, the first several layers of the network from the pretext task can be transferred to the downstream task. In the downstream task, only limited labeled samples are available. The downstream task is often not unique, and it may be classification, detection, or segmentation and so on. In the downstream task, the transferred layers are taken as the general feature extractor, and the remaining layers are finetuned by using labeled samples. The usage of effective feature representation layers from the pretext task can alleviate the overfitting and improve the performance of the downstream task.

III. Proposed SDC-GAN Method

The proposed SDC-GAN method is made up of two training processes. The first training process is for self-supervised

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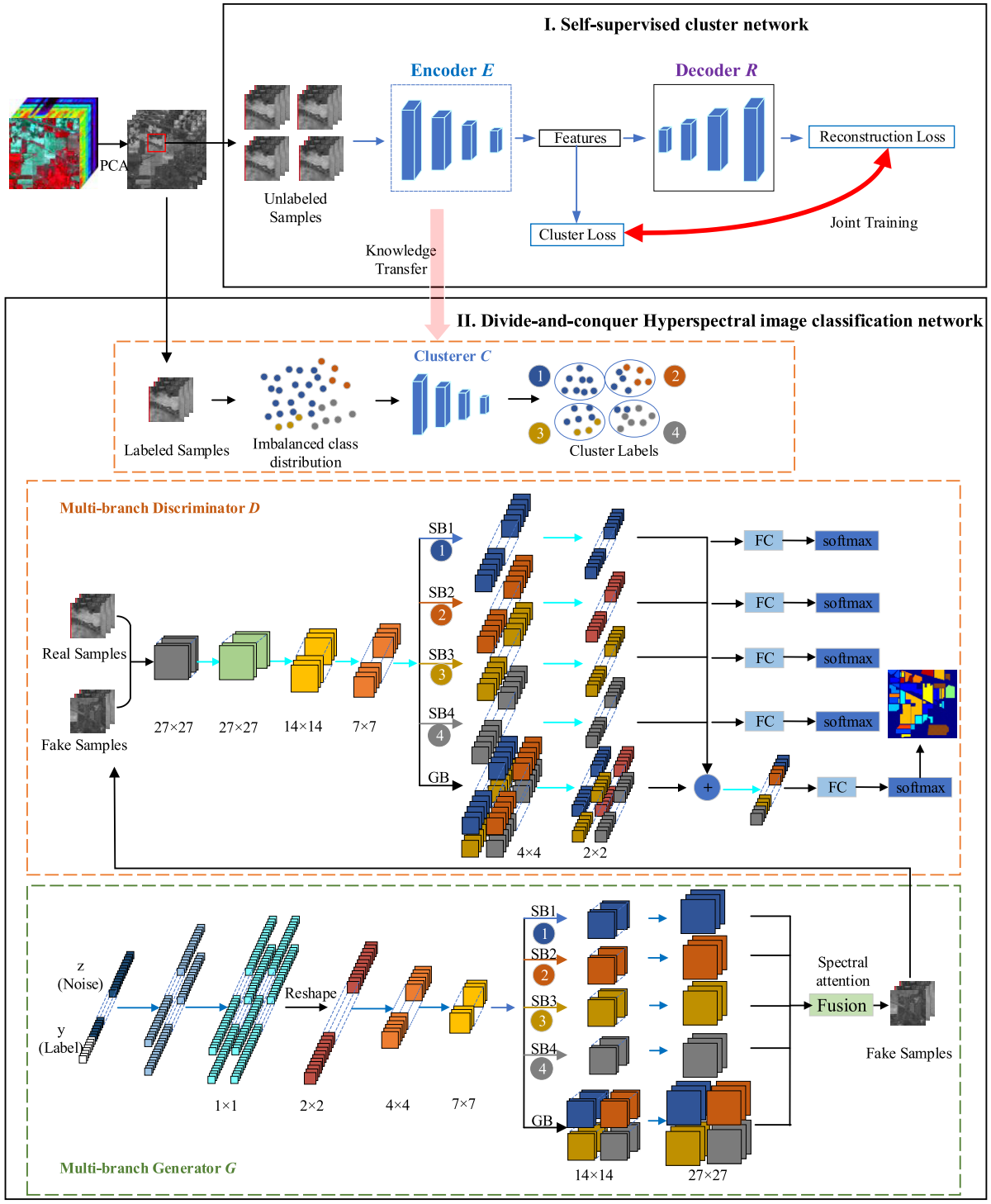


Fig. 3. Flowchart of the proposed SDC-GAN method. Clusterer C is transferred from the pretext task. Labeled training samples are divided into several clusters by clusterer C. Each cluster corresponds to one specific branch in generator G and discriminator D. SB and GB represent the specific and generic branches, respectively.

pretext cluster task and the second one is for HSI classification task. As shown in Fig. 3, the architecture of the self-supervised pretext cluster task is based on an encoder–decoder network, where more informative features can be extracted from amounts of unlabeled data. In the downstream HSI classification task, the idea of divide-and-conquer is used. This task mainly includes three networks: the clusterer C, the generator G, and the discriminator D. The clusterer C derives from the well-trained self-supervised pretext task, which has better clustering performance for limited labeled samples. Both the generator G and the discriminator D consist of a generic and several specific branches. The division of specific branches in G and D depends on the clustering results from the clusterer C. In G, compared with the generic branch, the number of

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classes is reduced and the training sample size in each class is relatively balanced in the specific branches. Then, the spectral attention mechanism is designed to combine the generation results from specific branches with the generic branch adaptively. In this way, G ameliorates the problem of imbalanced data distribution. In D, joint spectral–spatial features are extracted by cascading the 1-D and 2-D convolutional layers. Then, the similar hard samples will get into the customized specific branches to be further classified. By adding the auxiliary classifiers in specific branches and fusing the spectral–spatial features from all the branches, the classification ability of D is improved, especially for these hard samples.

A. Self-Supervised Pretext Cluster Task

In HSIs, $w \times w$ spatial neighborhood of each sample is used to represent this sample. The sample set is denoted as $X = \{x_i\}_{i=1}^{N}$. It is divided into three parts: labeled training samples $X^l = \{x_i\}_{i=1}^{N^l}$, unlabeled training samples $X^u = \{x_i\}_{i=1}^{N^u}$, and test samples $X^t = \{x_i\}_{i=1}^{N^t}$, where $N = N^l + N^u + N^t$. $Y^l = \{y_i\}_{i=1}^{N^l}$ is the corresponding label of X^l. In the self-supervised pretext cluster task, unlabeled samples X^u are used for training.

Inspired by Yang *et al.* [44], the self-supervised pretext cluster task is constructed by an encoder–decoder network. Amounts of unlabeled data are used as the input of the encoder. The high-level semantic features are extracted from the unlabeled data. Then, the unlabeled data are clustered through effective K-means method. The decoder reconstructs the input based on the semantic features. The clustering loss is used with the reconstruction loss to optimize the network, which can force the encoder to extract more representative and informative features for clustering. The loss function of self-supervised pretext cluster task L_c is calculated as follows:

$$L_c = \sum_{i=1}^{N^u} \left(\| R(E(x_i)) - x_i \|_2^2 + \| E(x_i) - M * s_i \|_2^2 \right) \quad (2)$$

where x_i represents the input sample from X^u, and E and R are the encoder and decoder networks. s_i is the clustering assignment of the ith sample, where only one value is 1 and all the other values are 0. M denotes the centroids of the clusters, "$*$" and "$||\cdot||$" represent the matrix multiplication and 2-norm, respectively.

The architecture of the encoder network includes four 2-D convolutional layers. The batch normalization (BN) layer and the rectified linear unit (ReLU) activation layer follow all the convolutional layers sequentially. Four transposed convolutional layers are constructed in the decoder network. The first three transposed convolutional layers are followed by BN and ReLU, and the last one uses Tanh. The detailed structures and parameters of the self-supervised cluster network are shown in Table I, where "BN?" represents whether BN is used.

Recently, some self-supervised learning methods have been applied to semantic segmentation tasks. These methods can be broadly classified into four classes: clustering-based methods [45], [46], Siamese network-based methods [47], contrastive learning-based methods [48], and other methods [49].

TABLE I
STRUCTURE AND PARAMETERS OF SELF-SUPERVISED CLUSTER NETWORK

Net	Convolution	BN?	Stride	Padding	Activation
E	3×3×16	Yes	2	1	ReLU
	3×3×32	Yes	2	1	ReLU
	3×3×64	Yes	2	1	ReLU
	3×3×128	No	2	1	ReLU

Net	Transposed Convolution	BN?	Stride	Padding	Activation
R	3×3×64	Yes	2	1	ReLU
	3×3×32	Yes	2	1	ReLU
	3×3×16	Yes	2	1	ReLU
	3×3×3	No	2	1	Tanh

In these cluster-based methods, their clustering objective is to give each pixel in the image a clustering label, where the number of clusters generally corresponds to the number of classes. However, the clustering objective of the proposed method is to roughly divide the input samples. By clustering the samples with high similarity into a cluster, it is prepared for the downstream classification task to deal with highly similar hard samples that are difficult to correctly distinguish. Compared with Siamese network-based methods, the proposed method uses a large number of unlabeled samples in the self-supervised pretext clustering task, which can improve the generalization ability of the downstream classification network. Compared with contrastive learning-based methods, the proposed method uses the GAN framework in the downstream task, and the multibranch generator can alleviate the problem of imbalanced class distribution, and the multibranch discriminator can improve the performance of hard samples with similar spectral features. All these designs can further improve the classification performance of HSIs.

B. Clusterer for Sample Division in SDC-GAN

In HSIs, limited labeled samples restrict the generating ability of GAN. The imbalanced class distribution in HSIs further aggravates the generating performance of the minority classes and easily causes the problem of mode collapse. The self-supervised pretext cluster network leverages amounts of unlabeled training data, which can learn more general and representative information and have powerful generalization ability for clustering. The clusterer C is obtained by transferring the learned feature representation from the self-supervised pretext cluster network, which will play a crucial role in subsequent missions.

The idea of divide-and-conquer is adopted to solve the HSI classification, where the "divide" is achieved by the clusterer C and the "conquer" is realized by the generator G and discriminator D. By the precise division of the clusterer C, the data distribution is relatively balanced than the original distribution and the similar hard samples would be grouped into the same cluster. In the generator G and discriminator D,

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the specific branches and a generic branch are constructed according to different divisions. The clusterer C provides the potential for the generator G to focus more on the samples of the minority classes and impels the discriminator D to learn fine-grained discriminative features for the hard samples.

Before the training of SDC-GAN, labeled training samples are divided into different clusters by using the clusterer C. After the clustering, each training sample will have a cluster label t_i, which represents the sequence of clusters that the sample belongs to

$$t_i = \arg\min_{1\le j\le k} \left\| E(x_i) - M_j \right\|_2^2, \quad x_i \in X^l. \tag{3}$$

As shown in (3), M_j represents the centroid of the jth cluster. $T = \{t_i\}_{i=1}^{N^l}$ is the cluster label of X^l. There are k clusters, which are denoted as $\{O_1, O_2, \ldots, O_k\}$.

In G, the random noise and the class labels of training samples are used as the inputs of the generator. After several layers, the obtained outputs are fed into the generic branch. Meanwhile, the outputs are divided into different specific branches according to the cluster labels, where the cluster label of the noise is obtained by assigning the cluster label corresponding to the training samples. The discriminator D takes real and generated samples as the inputs. The cluster labels of the generated samples correspond to that of the real samples. In D, joint spatial–spectral features extracted from the inputs will get into the generic branch and different specific branches according to the cluster labels.

C. Generator for Imbalanced Sample Generation in SDC-GAN

In the HSI classification task, the problem of imbalanced class distribution is very serious. For instance, the ratio of the sample size in the majority class to the minority class is as high as 120× in the Indian Pines dataset. In GAN-based classification methods, high-quality generated samples can assist to improve the classification performance of the discriminator. However, to fool the discriminator as much as possible, the generator would be biased toward learning the distribution of the samples of majority classes and undoubtedly ignore the distribution of minority classes, which will lead to the occurrence of mode collapse. Meanwhile, since HSIs contain abundant spectral and spatial information, it is difficult to guarantee that the generator generates high-quality samples in both spectral and spatial domains. These issues would further deteriorate the classification performance of the discriminator.

In the generator of SDC-GAN, specific branches and a generic branch are used to generate specific- and all-class samples individually. After the division of the clusterer C, different specific branches contain different classes, and the number of classes in the specific branch is much smaller than that in the generic branch. In addition, majority classes with larger sample sizes may input into different specific branches, so that the class distribution in the specific branches is more balanced than that in the generic branch. Then, each specific branch is responsible for generating specialized samples. In this way, SDC-GAN weakens the generation preference of the majority

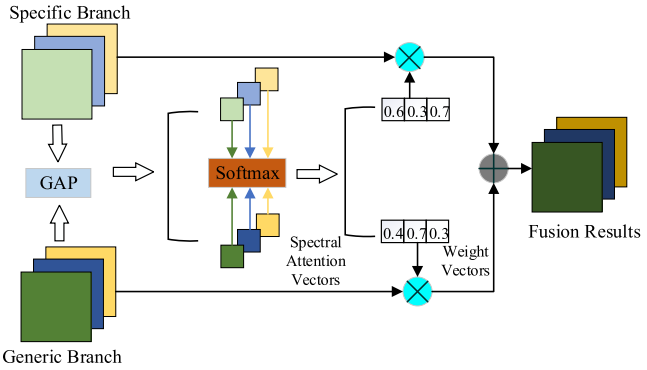


Fig. 4. Fusion process of the generator in SDC-GAN.

classes and mitigates the problem of imbalanced distribution effectively. Moreover, joint spectral–spatial information is generated by cascading the spectral and spatial transposed convolutional layers. In each branch, the spectral attention mechanism is designed to enhance the discriminative features and suppress features that are less important to the generated samples. Finally, high-quality samples can be obtained by fusing the results of these branches adaptively.

As shown in Fig. 3, two 1-D spectral transposed convolutional layers and two 2-D spatial transposed convolutional layers are followed by the inputs. According to the cluster labels mentioned-above, the obtained features would go into the generic branch and one of the specific branches. The generic and specific branches consist of two 2-D spatial transposed convolutional layers. After these two transposed convolutional layers, the spectral attention vectors are acquired through a global average pooling (GAP) layer. For each generated sample from the specific branch, we find its corresponding sample generated by the generic branch and then use the softmax function on these two spectral attention vectors to get the weight vectors. The final generated samples will be obtained by weighted summation of the samples from the specific branches and the general branch. Taking one specific branch as an example, Fig. 4 illustrates the fusion process of the samples generated by the specific and generic branch. The detailed design of G is shown in Table II. All the transposed convolutional layers are followed by BN and ReLU except the last one, where Tanh is utilized without BN. MS, SB, and GB denote the main stream network, the specific branch network, and the generic branch network, respectively.

D. Discriminator for Hard Sample Mining in SDC-GAN

HSI is a 3-D cube structure, which contains the quantities of spectral bands and has rich spatial information. However, influenced by light intensity, acquisition angle, and atmospheric humidity, the phenomenon of "different objects with similar spectrum" exists in HSIs. These highly similar samples are usually easy to be confused, which restricts the classification performance of HSIs. These samples are often called hard samples. How to effectively extract and distinguish these hard samples is focused here.

In the SDC-GAN, training samples will be divided into different clusters based on the similarity. After the division

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TABLE II
STRUCTURE AND PARAMETERS OF THE GENERATOR IN SDC-GAN

Net	Transposed Convolution	BN?	Stride	Padding	Activation
MS	3×2	Yes	1	1	ReLU
	3×4	Yes	1	1	ReLU
	3×3×64	Yes	2	1	ReLU
	3×3×32	Yes	2	1	ReLU
SB	3×3×16	Yes	2	1	ReLU
	3×3×3	No	2	1	Tanh
GB	3×3×16	Yes	2	1	ReLU
	3×3×3	No	2	1	Tanh

TABLE III
STRUCTURE AND PARAMETERS OF THE DISCRIMINATOR IN SDC-GAN

Net	Convolution	BN?	Stride	Padding	Activation
MS	3×16	Yes	1	1	ReLU
	3×32	Yes	1	1	ReLU
	3×3×64	Yes	2	1	ReLU
	3×3×128	Yes	2	1	ReLU
SB	3×3×256	Yes	2	1	ReLU
	3×3×512	No	2	1	ReLU
	FC	-	-	-	Softmax
GB	3×3×256	Yes	2	1	ReLU
	3×3×512	No	2	1	ReLU
	3×3×512	No	2	1	ReLU
	FC	-	-	-	Softmax

of the clusterer C, these hard samples have a high probability of belonging to the same cluster. In the discriminator D of SDC-GAN, the specific branches and a generic branch are constructed. Each specific branch is customized for the hard samples from the same cluster. In this way, the discriminator D uses specific branches to achieve fine-grained classification for these confusing hard samples, which can extract discriminative features and refine the final classification results.

To optimize the specific branches and the generic branch, a novel fused loss function is defined in SDC-GAN by combining the losses from these branches, which is calculated as follows:

$$L_D = L_{\text{gen}} + p\sum_{j=1}^{k} L_{\text{spe},j} \tag{4}$$

$$L_{\text{gen}} = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log(D(y|x))\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log D(y=c+1|G(z,y))\right] \tag{5}$$

$$L_{\text{spe},j} = \mathbb{E}_{\bar{x}\in O_j}\left[\log(D(\bar{y}|\bar{x}))\right] + \mathbb{E}_{z\sim p_z(z),\bar{y}\in Y_j}\left[\log D(\bar{y}=c+1|G(z,\bar{y}))\right] \tag{6}$$

$$L_G = \mathbb{E}_{z\sim p_z(z)}\left[\log D(y|G(z,y))\right] + p\sum_{j=1}^{k}\mathbb{E}_{z\sim p_z(z),\bar{y}\in Y_j}\left[\log D(\bar{y}|G(z,\bar{y}))\right] \tag{7}$$

where L_D and L_G represent the losses of the discriminator D and generator G, separately. L_{gen} and $L_{\text{spe},j}$ are the cross-entropy losses of the generic branch and the jth specific branch, respectively. Y_j is the class label of real samples in the jth cluster, and $Y^l = \{Y_1, Y_2, \ldots, Y_k\}$. $D(\cdot)$ and $G(\cdot)$ denote the outputs of D and G, and p is the coefficient that balances the losses from the generic and the specific branches.

D tries to maximize the loss of (4) by assigning the right labels to real samples and $c+1$ to generated samples, where c represents the number of classes. However, G wants to minimize the loss of (7) by dividing the generated samples into the first c classes. The distribution of generated samples is closer to that of the real samples by adversarial training, and the classification ability of D is improved gradually.

As shown in Fig. 3, the inputs of the discriminator D are real and generated samples. After 1-D spectral and 2-D spatial convolutional layers in D, the joint spectral–spatial features are extracted from the input. Based on the cluster labels, real and generated samples will get into the generic branch and one of the specific branches to extract discriminative features for classification. These specific branches consist of two convolutional layers, an FC layer and a softmax layer. After two convolutional layers in the generic branch, the obtained features are concatenated with these features before FC layer from the specific branches and then followed by a convolutional layer, an FC layer, and a softmax layer. The final classification map is achieved by the generic branch. The detailed design of D is shown in Table III.

E. Training Procedure of SDC-GAN

The training process of the proposed SDC-GAN method is shown in Table IV. The parameters of E and R, M, and s_i are updated in the training of the self-supervised pretext cluster task. The parameters of D and G are alternately updated in the classification task. Specifically, the parameters of D are updated when G is fixed and the parameters of G are updated by keeping D constant. To stabilize the training process of SDC-GAN, D is optimized once and G is optimized t times.

IV. EXPERIMENTAL RESULTS

In this section, three representative HS datasets are used to demonstrate the effectiveness of the proposed SDC-GAN algorithm. Several state-of-the-art HS classification methods are selected for comparison.

A. Data Description

The proposed SDC-GAN algorithm is carried out in the Indian Pines, Pavia University, and Washington datasets, and the detailed description of these datasets is shown in the following.

1) Indian Pines Dataset: This scene was gathered by the AVIRIS sensor over the Indian Pines test site in 1992. It contains 145 × 145 pixels with a spatial resolution of approximately 20 m and 200 spectral bands with the wavelengths ranging from 0.2 to 0.4 μm after 24 bands have been removed. This scene includes 16 categories. The false-color composite image and the ground truth of the Indian Pines dataset are shown in Fig. 5.

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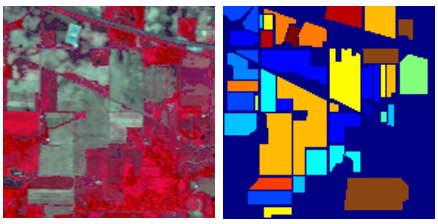
TABLE IV
DETAILED TRAINING PROCESS OF SDC-GAN

The training of self-supervised pretext cluster task

INPUT: the unlabeled training samples X^u, the number of training epochs E_1
Begin
Initialize the parameters of E, R and M, s_i,
for E_1 **epochs:**
  Update the parameters of networks E, R;
  Update the assignment s_i;
  Update the centroid M;
end for
End
OUTPUT: the well-trained E, M

The training of supervised downstream classification task

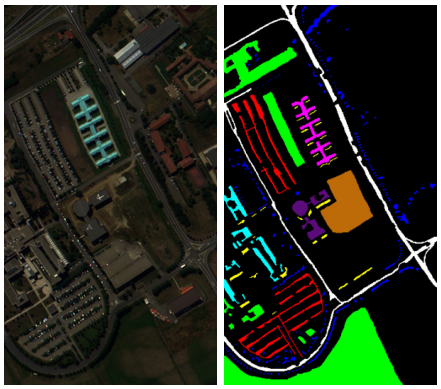
INPUT: the labeled training samples X^l, the labels Y of X^l, the test samples X^t, batch size m, the number of training epochs E_2, the updating times of the generator t
Begin
Initialize the parameters of D and G
for E_2 **epochs:**
  m random noise samples $z=\{z_1,z_2,...,z_m\}$ from Uniform distribution $U(-1,1)$ and m labeled samples
  concatenate the noises with corresponding labels $\{y_1,y_2,...,y_m\}$
  divide m labeled samples into k clusters with well-trained clusterer C by equation(3)
  generate fake samples $\{G(z_1,y_1),G(z_2,y_2),...,G(z_m,y_m)\}$
  update the parameters of D by maximizing the equation (7)
  for t **times:**
    update the parameters of G by minimizing the equation (8)
  end for
end for
End
OUTPUT: the predicted labels of test samples X^t by the well-trained D of SDC-GAN



(a) (b)

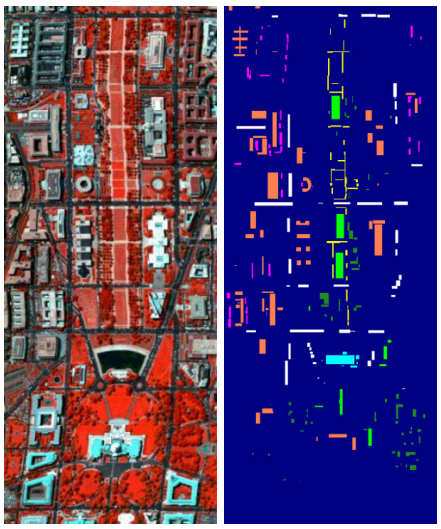
Fig. 5. Indian Pines dataset. (a) False-color composite image. (b) Ground truth.

2) Pavia University Dataset: This scene was captured by ROSIS-03 sensor over Pavia, northern Italy in 2003. It is composed of 610 × 345 pixels with a spatial resolution of 1.3 m and 115 spectral bands with the wavelengths ranging from 0.43 to 0.86 μm. After removing 12 noisy bands, 103 spectral bands are reduced and used in this article. This scene includes nine classes, and the false-color composite image and the ground truth of the Pavia University dataset are shown in Fig. 6.



(a) (b)

Fig. 6. Pavia University dataset. (a) False-color composite image. (b) Ground truth.



(a) (b)

Fig. 7. Washington dataset. (a) False-color composite image. (b) Ground truth.

3) Washington Dataset: This scene was collected by HYDICE sensor over the Washington dc mall in 1995. It consists of 750 × 307 pixels with a spatial resolution of 2.8 m and 191 spectral bands with the wavelengths ranging from 0.4 to 2.4 μm. This scene includes seven classes, and the false-color composite image and the ground truth of the Washington dataset are shown in Fig. 7.

B. Experimental Setting

To demonstrate the effectiveness of the SDC-GAN algorithm, several classical classification methods, including radial basis function (RBF)-SVM [9], semi-supervised (SS)-CNN [26], PPF-CNN [23], and 3D-CNN [19], and three GAN-based classification methods, HS-GAN [37], 3D-GAN [34], and AD-GAN [36], are utilized for comparison. SS-CNN and HS-GAN use the supplies of unlabeled samples to assist the classification. In this article, all the comparison methods will use their optimal parameters. For RBF-SVM, multiclassification

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is achieved by using the one-against-all strategy, and fivefold cross-validation is used to determine the penalty and gamma parameters. For PPF-CNN, the spatial neighborhood window size is set to the suggested value in [23]. For 3D-CNN, the spatial window size is set to 27×27. For HS-GAN, 1×3 and 1×5 convolutional kernels are used, and the number of training epochs is 200. For 3D-GAN, the 3-D input is $64 \times 64 \times 3$, where 64 is the spatial window size and three represents the dimensionality of channels. For the AD-GAN, the block size is set to 7 and the percentage k of discarded elements is set to 40, and the spatial window size is set to 27×27.

In SDC-GAN, the number of specific branches in both the generator and discriminator is set to four in the experiment. The number of training epochs is 300, and the learning rates of the generator and discriminator are 0.01 and 0.005. The parameter p that balances the generic and specific branches is set to 10 e-5, and the size of the spatial input window in the discriminator is 27×27. The updating time t of the generator is 1. Before the training of SDC-GAN, PCA is used to reduce the dimensionality of the input image and three principal components are retained, which decreases the training time and stabilizes the training process of SDC-GAN.

In the experiments, three indexes: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to validate the classification performance of the proposed SDC-GAN algorithm. All the experimental results displayed in this article are the averages of 20 independent experiments. All the experiments are carried out by using Python language and Pytorch library on an NVIDIA 2080Ti graphics card.

C. Classification Results of SDC-GAN

1) Classification Results on the Indian Pines Dataset: For the Indian Pines dataset, 10% samples of each class are randomly selected as the unlabeled training set to train the pretext cluster network, 5% to train the SDC-GAN, and the remaining 85% is used for testing. Table V shows the detailed numbers of different classes. The quantitative results of SDC-GAN and the other seven comparison algorithms on the Indian Pines dataset are shown in Table VI, including OA, AA, Kappa, and the accuracy of each category. Among these experimental results, the highest accuracy is highlighted with a gray background color.

As shown in Table VI, compared with RBF-SVM, deep learning-based algorithms, SS-CNN, PPF-CNN, 3D-CNN, 3D-GAN, AD-GAN, and SDC-GAN, achieve higher accuracy because of the extraction of discriminant high-level semantic features. Extra unlabeled samples are used in SS-CNN, which improves the OA index by 1.4% than RBF-SVM. A large number of expanded samples improve the accuracy of PPF-CNN, and its OA index is 8.7% higher than that of SS-CNN. 3D-CNN takes advantage of joint spatial–spectral information, which has a better result than PPF-CNN. Although HS-GAN uses sufficient unlabeled samples, it only extracts spectral features. 3D-GAN and AD-GAN generate 3-D spatial–spectral samples containing more information of real samples, and their classification results are superior to HS-GAN. Among the eight

TABLE V
NUMBERS OF LABELED, UNLABELED, AND TEST SAMPLES ON THE INDIAN PINES DATASET

Class		Numbers of samples		
No	Name	Labeled	Unlabeled	Test
1	Alfalfa	2	5	39
2	Corn-notill	71	143	1214
3	Corn-mintill	42	83	705
4	Corn	12	24	201
5	Grass-pasture	24	48	411
6	Grass-trees	36	73	621
7	Grass-pasture-mowed	1	3	24
8	Hay-windrowed	24	48	406
9	Oats	1	2	17
10	Soybean-notill	49	97	826
11	Soybean-mintill	123	246	2086
12	Soybean-clean	30	59	504
13	Wheat	10	20	175
14	Woods	63	126	1076
15	Buildings-Grass-Trees-Drives	19	39	328
16	Stone-Steel-Towers	5	9	79
Total		512	1025	8712

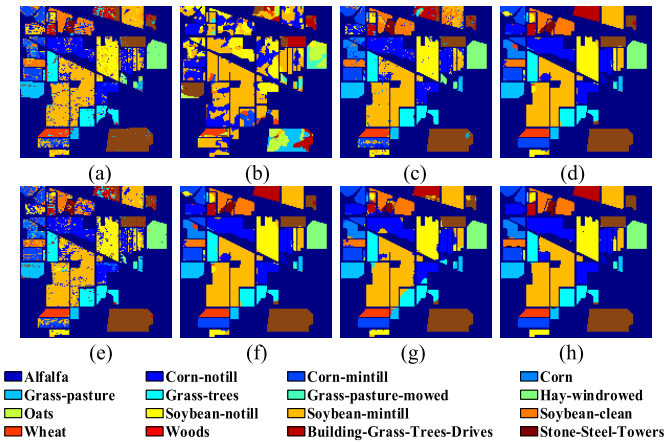


Fig. 8. Classification visual maps on the Indian Pines dataset. (a) RBF-SVM, (b) SS-CNN, (c) PPF-CNN, (d) 3D-CNN, (e) HS-GAN, (f) 3D-GAN, (g) AD-GAN, and (h) SDC-GAN.

methods, SDC-GAN achieves the best performance in most of the classes, especially in minority classes, such as Alfalfa and Oats. Compared with other methods, SDC-GAN improves the classification accuracy in the OA, AA, and Kappa by at least 2.8%, 14.5%, and 3.2%, separately.

The visual diagrams of the classification results on the Indian Pines dataset are displayed in Fig. 8. As shown in Fig. 8, RBF-SVM, SS-CNN, PPF-CNN, and HS-GAN have many wrong classification noise points in some regions of majority classes, such as Corn-notill, Soybean-notill, and Soybean-mintill. AD-GAN does not perform well in the classification of Alfalfa, Grass-pasture-mowed, and Oats. Compared with these methods, 3D-CNN, 3D-GAN, and SDC-GAN have better visual maps and improve the region uniformity. Specifically, SDC-GAN has the best visual effect in Oats.

2) Classification Results on the Pavia University Dataset: For the Pavia University dataset, we randomly choose 6% of each class as the unlabeled training set to train the pretext

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TABLE VI
CLASSIFICATION RESULTS ON THE INDIAN PINES DATASET

Class	RBF-SVM[9]	SS-CNN[26]	PPF-CNN[23]	3D-CNN[19]	HS-GAN[37]	3D-GAN[34]	AD-GAN[36]	SDC-GAN
1	6.1±11.2	35.3±25.1	50.4±8.4	83.8±13.4	17.7±5.2	90.9±5.2	47.7±0.7	98.9±2.1
2	72.9±3.6	76.8±21.7	89.2±2.1	92.7±3.5	66.3±1.1	91.0±1.7	91.1±0.2	95.6±0.3
3	58.0±3.6	60.1±20.4	77.1±2.7	87.2±10.4	60.2±2.9	90.4±2.1	98.9±0.2	96.5±1.2
4	39.0±15.0	67.4±11.9	87.7±3.7	83.4±8.3	57.8±4.7	93.7±4.3	91.6±0.6	98.9±0.7
5	87.0±4.5	90.6±8.6	92.7±1.0	84.0±5.7	82.0±6.1	93.2±4.5	95.2±0.2	93.7±0.7
6	92.4±2.0	98.8±1.7	93.1±1.9	93.4±2.5	94.3±2.2	95.4±0.7	92.7±0.3	99.0±0.4
7	0±0	43.8±43.8	0±0	97.2±4.8	23.8±12.2	94.9±0.1	29.6±1.1	87.0±5.5
8	98.1±1.4	94.9±18.5	99.6±0.3	97.4±2.8	98.8±0.3	99.9±0.2	99.8±0.1	99.9±0.1
9	0±0	67.9±15.2	0±0	77.0±11.1	13.7±12.1	53.5±1.4	21.0±0.7	95.3±6.8
10	65.8±3.7	60.2±57.9	85.6±2.8	93.3±5.0	68.5±3.6	94.2±0.3	87.1±0.3	95.9±0.9
11	85.3±2.9	90.6±13.1	83.8±1.6	94.9±2.7	79.7±0.5	94.7±1.5	97.8±0.1	99.5±0.3
12	69.6±6.5	52.1±33.2	90.4±3.1	89.8±4.3	48.8±4.5	92.1±2.3	90.9±0.5	90.0±1.1
13	92.3±4.1	99.8±0.8	97.8±0.9	92.8±5.9	89.2±2.7	95.5±0.2	100.0±0.1	98.5±1.3
14	96.6±1.0	95.0±4.6	95.5±1.1	98.3±1.3	96.0±1.1	95.6±0.3	97.5±0.1	95.0±0.4
15	41.7±7.0	26.2±25.9	78.0±2.4	77.8±13.4	37.9±11.4	87.7±2.1	94.8±0.7	98.3±1.1
16	75.2±9.0	89.8±42.1	97.3±1.3	88.4±5.3	73.0±5.3	92.6±2.3	67.0±0.5	91.2±1.3
OA(%)	77.8±0.8	79.2±2.7	87.9±0.8	92.8±0.8	74.0±0.9	93.5±0.3	94.0±0.5	96.8±0.2
AA(%)	61.3±1.4	71.8±1.1	76.5±0.6	89.4±1.4	60.2±2.6	84.8±2.7	81.4±1.0	95.9±0.8
Kappa(%)	74.5±1.0	75.8±4.5	86.3±0.9	91.9±0.9	70.0±1.0	93.1±1.2	93.1±0.6	96.3±0.2

TABLE VII
NUMBERS OF LABELED, UNLABELED, AND TEST SAMPLES ON THE PAVIA UNIVERSITY DATASET

Class		Numbers of samples		
No	Name	Labeled	Unlabeled	Test
1	Asphalt	199	398	6034
2	Meadows	559	1119	16971
3	Gravel	63	126	1910
4	Trees	92	184	2788
5	Painted metal sheets	40	81	1224
6	Bare Soil	151	302	4576
7	Bitumen	40	80	1210
8	Self-Blocking Bricks	110	221	3351
9	Shadows	28	57	862
Total		1282	2568	38926

cluster network, 3% to train the SDC-GAN, and the remaining 91% is used for testing. Table VII lists the number of samples for different classes. The quantitative results of all the algorithms on the Pavia University dataset are shown in Table VIII.

As shown in Table VIII, the OA indexes of SS-CNN and HS-GAN are below 90%, and the OA of SS-CNN is the lowest. HS-GAN improves the OA index by 0.3% than SS-CNN. The OA indexes of RBF-SVM, PPF-CNN, 3D-CNN, 3D-GAN, AD-GAN, and SDC-GAN are above 90%, which are 4.9%, 11.5%, 9.8%, 12.8%, 12.9%, and 13.7% higher than that of SS-CNN, respectively. PPF-CNN has a better result than SS-CNN and 3D-CNN. Compared with HS-GAN and 3D-GAN, AD-GAN obtains the highest OA index, followed by 3D-GAN and HS-GAN. Among the eight methods, SDC-GAN achieves the best performance in most of the classes, such as Meadows and Painted metal sheets. Compared with other methods, SDC-GAN improves the classification accuracy in the OA, AA, and Kappa indexes by at least 0.8%, 1.1%, and 1.1%, separately.

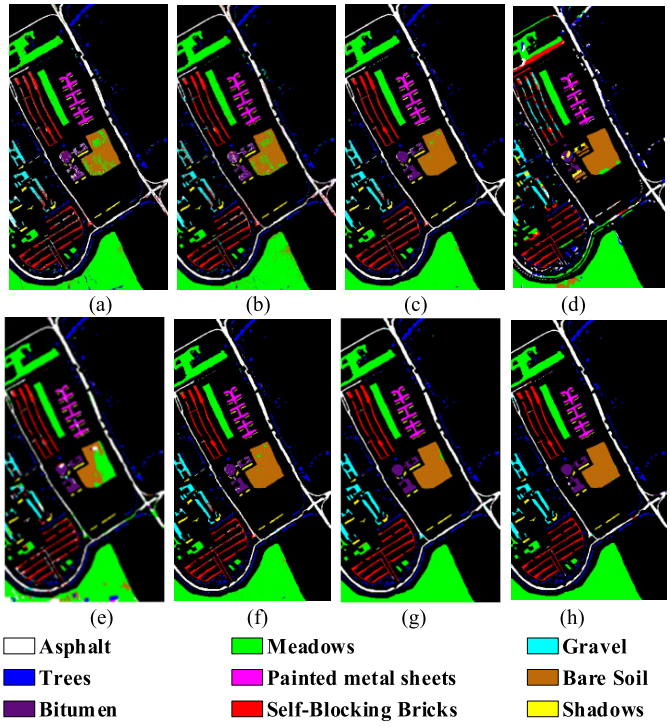


Fig. 9. Classification visual maps on the Pavia University dataset. (a) RBF-SVM, (b) SS-CNN, (c) PPF-CNN, (d) 3D-CNN, (e) HS-GAN, (f) 3D-GAN, (g) AD-GAN, and (h) SDC-GAN.

The visual diagrams of the classification results on the Pavia University dataset are displayed in Fig. 9. As shown in Fig. 9,

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TABLE VIII
CLASSIFICATION RESULTS ON THE PAVIA UNIVERSITY DATASET

Class	RBF-SVM[9]	SS-CNN[26]	PPF-CNN[23]	3D-CNN[19]	HS-GAN[37]	3D-GAN[34]	AD-GAN[36]	SDC-GAN
1	90.7±1.1	60.7±38.3	98.0±0.1	95.5±1.2	87.4±0.9	99.3±0.4	99.3±0.1	99.6±0.2
2	96.8±0.7	98.4±1.5	99.2±0.2	99.4±0.3	93.0±1.9	99.7±0.3	99.7±0.0	99.9±0.1
3	60.2±5.4	96.4±4.6	84.9±1.8	92.6±5.4	63.6±7.2	92.5±2.3	94.1±0.1	96.6±0.8
4	90.8±2.0	93.9±1.9	95.8±0.8	75.2±4.9	89.6±4.0	93.2±2.3	97.3±0.1	98.0±0.9
5	98.8±0.4	40.2±40.2	99.8±0.1	95.4±4.3	99.6±0.2	99.8±0.3	99.9±0.2	99.9±0.1
6	79.5±4.9	94.2±3.3	96.4±0.3	99.4±0.6	70.5±7.8	98.7±0.8	94.6±0.1	99.5±0.1
7	74.3±5.1	99.0±2.0	89.2±0.8	91.5±3.4	69.7±6.0	92.2±3.3	94.7±0.7	95.0±1.9
8	88.8±2.2	47.1±23.9	93.7±1.2	94.8±1.4	73.6±4.3	98.0±1.2	98.4±0.2	99.0±0.3
9	99.8±0.1	98.7±2.4	98.5±0.7	77.4±2.8	97.8±1.3	94.5±3.7	96.8±0.6	97.9±1.2
OA(%)	90.3±0.6	85.4±2.4	96.9±0.2	95.2±0.7	85.7±0.2	98.2±0.1	98.3±0.0	99.1±0.3
AA(%)	86.6±0.9	81.0±1.0	95.1±0.2	91.2±1.1	82.8±0.8	96.4±0.5	97.2±0.1	98.3±0.5
Kappa(%)	87.1±0.8	80.9±3.2	96.0±0.2	93.8±0.9	81.0±0.3	97.6±0.2	97.7±0.0	98.8±0.4

TABLE IX
NUMBERS OF LABELED, UNLABELED, AND TEST SAMPLES ON THE WASHINGTON DATASET

Class		Numbers of samples		
No	Name	Labeled	Unlabeled	Test
1	Roads	86	172	2615
2	Grass	51	103	1560
3	Water	19	38	573
4	Roofs	31	62	943
5	Trails	38	77	1163
6	Trees	35	69	1049
7	Shadows	168	337	5106
Total		428	858	13009

RBF-SVM has poor classification performance in the gravel, bare soil, and bitumen categories. SS-CNN misclassifies many samples in asphalt, painted metal sheets, and self-blocking bricks. 3D-CNN does not perform well in the classification of trees and shadows. HS-GAN does not classify gravel, bare soil, and self-blocking bricks well. Compared with these methods, SDC-GAN has the best visual map, where the boundaries are clearer and more accurate, and regional consistency is better maintained.

3) Classification Results on the Washington Dataset: For the Washington dataset, we randomly pick 6% of each class as the unlabeled training set to train the pretext cluster network, 3% to train the SDC-GAN, and the remaining 91% is used for testing. The numbers of training and test samples for each class are listed in Table IX. The quantitative results of all the algorithms on the Washington dataset are shown in Table X.

As shown in Table X, the OA indexes of all the methods are above 90%. However, RBF-SVM has a poor classification performance in Roofs. SS-CNN misclassifies many samples in roads. 3D-CNN does not perform well in trails. Compared with RBF-SVM and CNN-based methods, 3D-GAN, AD-GAN, and SDC-GAN achieve higher accuracy except for HS-GAN. The OA index of SS-CNN is 0.3% and 1.6% higher than PPF-CNN and 3D-CNN. Compared with HS-GAN and 3D-GAN, AD-GAN achieves the best OA, which is 5.0% and 0.3% higher than HS-GAN and 3D-GAN. Among these methods, SDC-GAN shows the best classification accuracy, which improves the OA, AA, and Kappa by at least 2.0%, 2.2%, and 2.5%, separately.

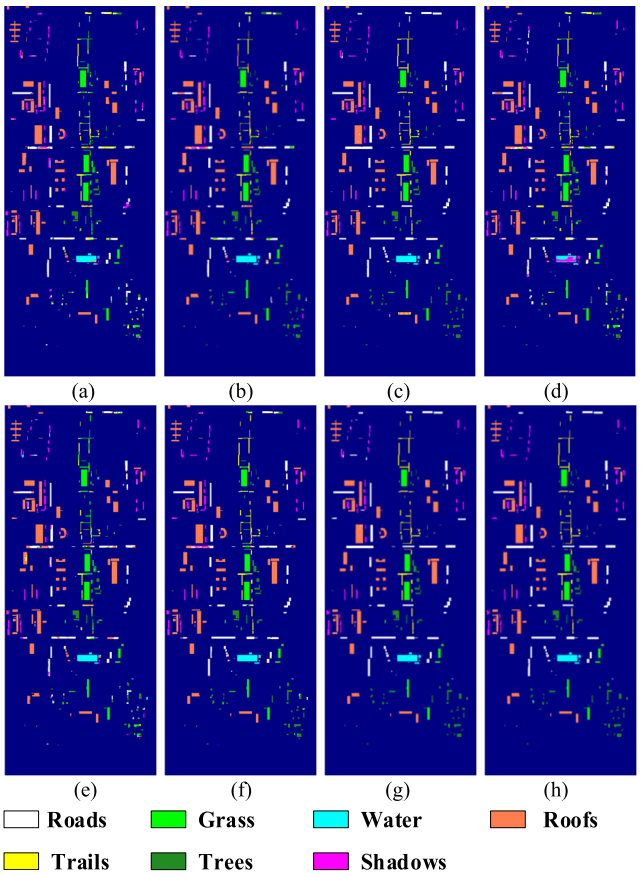


Fig. 10. Classification visual maps on the Washington dataset. (a) RBF-SVM, (b) SS-CNN, (c) PPF-CNN, (d) 3D-CNN, (e) HS-GAN, (f) 3D-GAN, (g) AD-GAN, and (h) SDC-GAN.

The visual diagrams of the classification results on the Washington dataset are displayed in Fig. 10. As shown in

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TABLE X
CLASSIFICATION RESULTS ON THE WASHINGTON DATASET

Class	RBF-SVM[9]	SS-CNN[26]	PPF-CNN[23]	3D-CNN[19]	HS-GAN[37]	3D-GAN[34]	AD-GAN[36]	SDC-GAN
1	94.1±3.1	89.2±13.1	97.9±0.6	92.5±2.2	92.8±3.1	96.1±0.1	95.5±1.4	99.4±0.1
2	93.4±0.6	99.9±0.2	97.6±0.1	92.8±7.0	94.9±0.1	95.4±3.8	99.2±0.6	100.0±0.0
3	98.3±0.1	99.8±0.3	100.0±0.0	91.6±5.8	95.8±0.3	99.6±0.0	99.6±0.6	100.0±0.0
4	88.2±3.9	92.8±3.5	95.6±3.1	91.5±3.2	90.8±3.5	99.0±1.1	94.4±3.8	99.3±0.1
5	95.6±0.4	100.0±0.0	99.9±0.1	88.0±5.2	90.0±0.0	99.5±0.3	98.2±0.5	99.2±0.0
6	91.6±3.5	90.6±11.1	97.5±1.2	90.4±2.4	91.6±1.1	97.0±1.0	89.8±3.5	96.4±0.0
7	98.2±1.5	98.7±1.3	94.9±0.1	99.0±0.4	94.7±1.3	98.2±0.7	99.8±0.3	100.0±0.0
OA(%)	93.7±0.4	96.0±3.6	95.7±0.3	94.4±0.6	92.5±1.6	97.2±0.3	97.5±0.5	99.5±0.0
AA(%)	92.3±0.8	95.8±3.8	95.9±0.5	92.3±0.8	90.8±1.8	97.0±0.5	96.7±0.6	99.2±0.0
Kappa(%)	93.7±0.6	94.8±4.6	95.5±0.3	92.8±0.7	90.3±1.6	96.7±0.4	96.8±0.6	99.3±0.0

TABLE XI
RUNNING TIME ON THE INDIAN PINES DATASET

Dataset	Method	Training Time(s)	Test Time(s)
Indian Pines	RBF-SVM[9]	0.4±0.1	1.2±0.1
	SS-CNN[25]	178.8±3.1	0.2±0.1
	PPF-CNN[22]	2056.0±36.7	5.3±0.3
	3D-CNN[19]	2690.2±57.9	16.0±0.1
	HS-GAN[32]	444.7±73.1	0.3±0.0
	3D-GAN[29]	597.67±60.8	0.3±0.0
	AD-GAN[31]	327.4±4.2	1.4±0.0
	SDC-GAN	497.6±2.0	0.8±0.0

TABLE XII
RUNNING TIME ON THE PAVIA UNIVERSITY DATASET

Dataset	Method	Training Time(s)	Test Time(s)
Pavia University	RBF-SVM[9]	0.5±0.1	1.4±0.2
	SS-CNN[25]	209.9±0.9	0.3±0.0
	PPF-CNN[22]	2414.0±374.0	19.8±6.2
	3D-CNN[19]	1979.0±12.6	31.4±5.5
	HS-GAN[32]	580.2±20.5	0.5±0.1
	3D-GAN[29]	724.4±50.7	0.6±0.1
	AD-GAN[31]	694.6±26.8	2.9±0.0
	SDC-GAN	712.2±5.9	2.1±0.4

TABLE XIII
RUNNING TIME ON THE WASHINGTON DATASET

Dataset	Method	Training Time(s)	Test Time(s)
Washington	RBF-SVM[9]	0.3±0.0	0.2±0.0
	SS-CNN[25]	80.6±0.5	0.1±0.0
	PPF-CNN[22]	926.8±29.5	5.2±0.5
	3D-CNN[19]	1588.8±18.6	7.8±0.2
	HS-GAN[32]	493.4±73.8	0.2±0.1
	3D-GAN[29]	673.3±23.7	0.3±0.1
	AD-GAN[31]	301.3±5.3	1.9±0.0
	SDC-GAN	632.4±1.8	1.0±0.1

Fig. 10, RBF-SVM, SS-CNN, 3D-CNN, and HS-GAN have many misclassified regions in roads and trails. PPF-CNN, AD-GAN, and SDC-GAN have better classification results and region uniformity in these two classes. HS-GAN and AD-GAN misclassify many samples in roofs. Compared with these methods, SDC-GAN has the best visual maps, and the boundaries are clearer and more accurate. In addition, there are few misclassified points within the region.

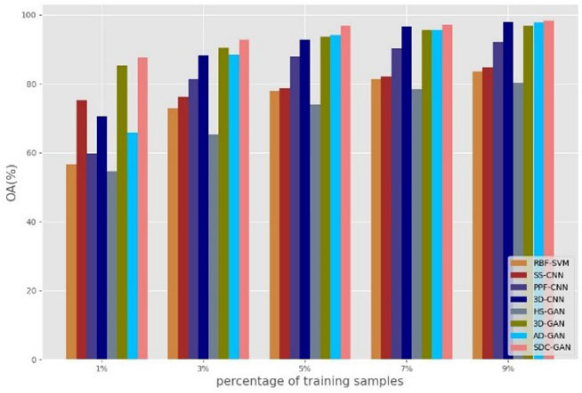
D. Analysis on Running Time

The training and test time on the three HSI datasets are recorded in Tables XI–XIII. In the training phase, RBF-SVM takes the least time because the inputs are 1-D spectral vectors. In the CNN-based methods, PPF-CNN and 3D-CNN consume more time than SS-CNN. The expanded training samples increase the training time of PPF-CNN. The 3-D patches used in 3D-CNN need the calculation of 3-D convolution, which increases the number of parameters and computation. In GAN-based methods, 3D-GAN and SDC-GAN take more time than HS-GAN and AD-GAN. 3D-GAN uses a larger size of input patch than AD-GAN and SDC-GAN. SDC-GAN requires more time for the clustering division and the training of multiple branches. In the test phase, PPF-CNN and 3D-CNN consume more time because of the vote strategy and 3-D convolution. AD-GAN and SDC-GAN cost more time than RBF-SVM, SS-CNN, HS-GAN, and 3D-GAN, but they take less time than PPF-CNN and 3D-CNN.

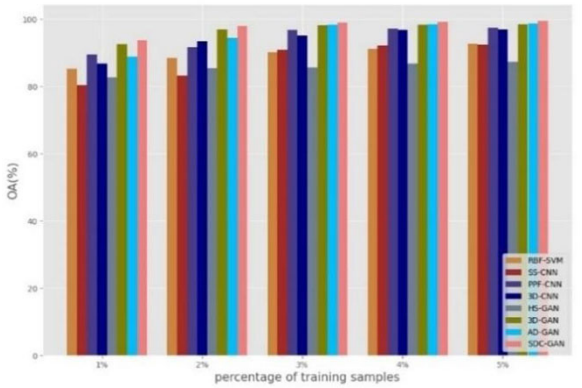
E. Sensitivity to the Number of Training Samples

In this part, different proportions of training samples are utilized to validate the effectiveness of the eight methods. As displayed in Fig. 11(a)–(c), 1%, 3%, 5%, 7%, and 9% labeled training samples in each class are used on the Indian Pines dataset; 1%, 2%, 3%, 4%, and 5% training samples in each class are used on the Pavia University and Washington datasets. With the increasing proportion of training samples, the classification accuracy of each method is improved, but the speed of the improvement slows down gradually and tends to be stable. On the Indian Pines and Pavia University datasets, 3D-GAN performs better than AD-GAN in the sample proportion of 1% and 2% but worse than AD-GAN in the sample proportion of 3%, 4%, and 5%. On the Washington dataset, AD-GAN is superior to 3D-GAN in all the cases. Among these methods, SDC-GAN achieves the highest accuracy in each proportion, especially with 1% training samples. It improves the OA index by at least 2.4%, 1.2%, and 0.6% on the Indian Pines, Pavia University, and Washington datasets, respectively.

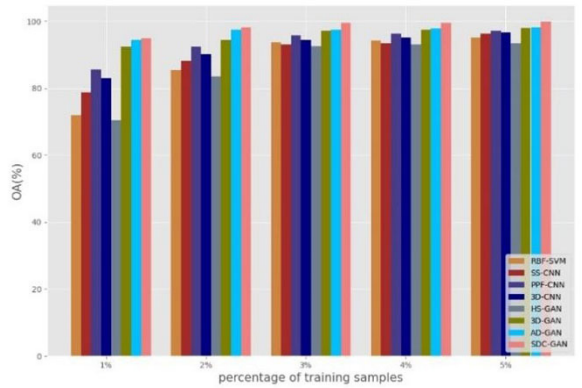
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(a)



(b)



(c)

Fig. 11. OA results with different percentages of training samples. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Washington dataset.

SDC-GAN shows the potential to deal with the problem of small sample size.

F. Structural Validity Analysis of SDC-GAN

To verify the effectiveness of self-supervised learning, the divide-and-conquer GAN with random clustering (RDC-GAN) is used for comparison. In RDC-GAN, self-supervised learning with amounts of unlabeled data is deleted, and a random clustering strategy is used for labeled sample division in both the generator and discriminator. To demonstrate the effectiveness of each component in GAN of the SDC-GAN algorithm, SDC-GAN with only one generic branch (GAN), SDC-GAN with multibranch generator (SMG-GAN), and SDC-GAN with multibranch discriminator (SMD-GAN) are used as comparison methods.

TABLE XIV
CLASSIFICATION RESULTS OF SDC-GAN AND RDC-GAN ON THE THREE DATASETS

Dataset	Method	OA(%)	AA(%)	Kappa(%)
Indian Pines	SDC-GAN	96.8±0.2	95.9±0.8	96.3±0.2
	RDC-GAN	93.3±0.1	88.5±0.4	92.3±0.2
Pavia University	SDC-GAN	99.1±0.3	98.3±0.5	98.8±0.4
	RDC-GAN	98.3±0.2	97.3±0.4	97.8±0.1
Washington	SDC-GAN	99.5±0.0	99.2±0.0	99.3±0.0
	RDC-GAN	98.8±0.1	98.3±0.2	98.5±0.2

TABLE XV
CLASSIFICATION RESULTS OF GAN, SMG-GAN, SMD-GAN, AND SDC-GAN ON THE THREE DATASETS

Dataset	Method	OA(%)	AA(%)	Kappa(%)
Indian Pines	GAN	89.8±0.2	80.3±0.2	88.5±0.2
	SMG-GAN	92.4±0.1	84.8±0.2	91.3±0.1
	SMD-GAN	92.3±0.2	87.2±0.2	91.3±0.2
	SDC-GAN	96.8±0.2	95.9±0.8	96.3±0.2
Pavia University	GAN	94.2±0.2	90.0±0.3	92.4±0.2
	SMG-GAN	98.7±0.1	97.8±0.1	98.2±0.1
	SMD-GAN	98.9±0.1	98.2±0.0	98.6±0.1
	SDC-GAN	99.1±0.3	98.3±0.5	98.8±0.4
Washington	GAN	96.4±0.5	95.3±1.2	95.3±0.6
	SMG-GAN	99.0±0.1	98.7±0.1	98.7±0.2
	SMD-GAN	99.0±0.2	98.8±0.2	98.6±0.2
	SDC-GAN	99.5±0.0	99.2±0.0	99.3±0.0

Table XIV lists the statistical results of SDC-GAN and RDC-GAN. SDC-GAN outperforms RDC-GAN on three HSI datasets because training samples can be precisely divided by self-supervised pretext cluster task. After partitioning, the specific branches in G of SDC-GAN can better alleviate the problem of generation preference of majority classes. The specific branches in D of SDC-GAN have stronger pertinence for the classification of hard samples. The classification performance in Table XIV proves the superiority of self-supervised learning.

In Table XV, GAN has the lowest results in each dataset. There is little difference in classification accuracy between SMG-GAN and SMD-GAN, and SDC-GAN obtains the highest classification accuracies. Compared with GAN, SMG-GAN improves the OA index by 2.6%, 4.5%, and 2.6% on the Indian Pines, Pavia University, and Washington datasets, separately. It verifies that the usage of specific branches in both G and D is beneficial for classification. Compared with SMD-GAN, SDC-GAN improves the OA index by 4.5%, 0.2%, and 0.5% on the three datasets, which demonstrates that the combination of multibranch G and D can further improve the classification performance.

G. Visualization of Generated Samples by SDC-GAN

As shown in Fig. 12, the generated samples of the category with the smallest sample size on the three datasets are

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TABLE XVI
CLASSIFICATION RESULTS WITH DIFFERENT UNLABELED SAMPLES ON THE THREE DATASETS

Dataset	Index	1:1	2:1	3:1	4:1	5:1
Indian Pines	OA(%)	96.0±0.3	96.8±0.2	96.9±0.3	96.8±0.1	96.9±0.2
	AA(%)	95.1±0.4	95.9±0.8	96.2±0.5	96.0±0.4	96.3±0.5
	Kappa(%)	95.5±0.4	96.3±0.2	96.5±0.2	96.3±0.5	96.5±0.3
Pavia University	OA(%)	98.9±0.1	99.1±0.3	99.1±0.1	99.2±0.2	99.1±0.1
	AA(%)	98.1±0.2	98.3±0.5	98.2±0.3	98.5±0.4	98.4±0.4
	Kappa(%)	98.6±0.1	98.8±0.4	98.8±0.1	99.0±0.0	98.8±0.2
Washington	OA(%)	99.1±0.2	99.5±0.0	99.5±0.2	99.6±0.1	99.5±0.1
	AA(%)	99.2±0.3	99.2±0.0	99.2±0.6	99.1±0.3	99.3±0.1
	Kappa(%)	98.9±0.3	99.3±0.0	99.4±0.3	99.3±0.1	99.4±0.1

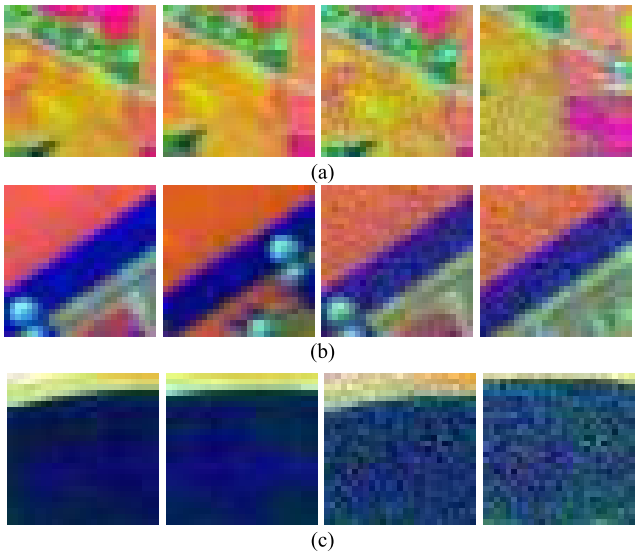


Fig. 12. Visualization of generated samples. From left to right: real samples and generated samples by SDC-GAN, AD-GAN, and 3D-GAN. (a) Indian Pines dataset: Oats. (b) Pavia University dataset: Shadows. (c) Washington dataset: Water.

selected for display. From left to right are the real samples and the samples generated by SDC-GAN, AD-GAN, and 3D-GAN, respectively. The sample size generated by 3D-GAN is 64 × 64. In order to make a fair comparison with AD-GAN and SDC-GAN, the central area with the size of 27 × 27 is cut for visualization.

We can see that the samples generated by SDC-GAN are similar to the real samples but not the same, which have the best visual effect. The samples generated by AD-GAN are similar to the real samples but contain more noise. 3D-GAN can learn some macroscopic features, but it is easy to ignore the generation of some details, and it also contains much noise. In other words, after finishing the training of the SDC-GAN, the generator G can generate diverse expanded samples. It is beneficial for the discriminator D to learn richer features and improve classification accuracy, especially in the case of limited training samples.

H. Analysis of Different Proportions of Unlabeled Samples to Labeled Samples

In this part, different proportions of unlabeled samples to labeled samples are used to verify the influence of unlabeled samples on the proposed SDC-GAN method. The experimental results on three HSI datasets are listed in Table XVI.

As shown in Table XVI, when the proportion of unlabeled samples to labeled samples is changed from 1:1 to 2:1, the OA, AA, and Kappa indices on the three datasets are improved significantly. With the further increase of the proportion, these results do not improve much and tend to be stable overall. In general, more unlabeled samples can get a good clustering result, which can improve the classification performance. However, the effect of too many unlabeled samples on improving the classification accuracy is not obvious. The reason maybe is that there are enough unlabeled samples with the proportion of 2:1 to provide a good clustering result. Meanwhile, too many unlabeled samples will cause a waste of computing resources.

TABLE XVII
CLASSIFICATION RESULTS OF MDGCN, SSTN, AND SDC-GAN ON THE THREE DATASETS

Dataset	Index	MDGCN[50]	SSTN[51]	SDC-GAN
Indian Pines	OA(%)	94.8±0.1	95.1±0.3	96.8±0.2
	AA(%)	79.5±1.9	92.1±0.2	95.9±0.8
	Kappa(%)	95.3±0.8	94.4±0.1	96.3±0.2
Pavia University	OA(%)	97.2±0.1	98.1±0.2	99.1±0.3
	AA(%)	91.7±0.3	97.1±0.3	98.3±0.5
	Kappa(%)	96.3±0.1	97.5±0.1	98.8±0.4
Washington	OA(%)	97.4±0.1	98.3±0.1	99.5±0.0
	AA(%)	96.7±0.1	98.0±0.2	99.2±0.0
	Kappa(%)	96.6±0.1	97.8±0.2	99.3±0.0

I. Comparison With Other Deep Learning Algorithms

To verify the superiority of the method proposed in the article, this part compares with a GCN-based method, MDGCN [50], and a transformer-based method, SSTN [51], respectively. The experimental results on three HSI datasets are listed in Table XVII.

As shown in Table XVII, SSTN achieves better classification performance than MDGCN by capturing long-range interactions with more flexible transformer units and exceeds MDGCN by 0.3%, 0.9%, and 0.9% in terms of OA on three HSI datasets. Compared with MDGCN and SSTN, the proposed SDC-GAN method achieves better classification performance on three HSI datasets.

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TABLE XVIII
CLASSIFICATION RESULTS ON THE VAIHINGEN DATASET

Index	RBF-SVM[9]	SS-CNN[26]	PPF-CNN[23]	3D-CNN[19]	HS-GAN[37]	3D-GAN[34]	AD-GAN[36]	SDC-GAN
OA(%)	62.8±0.1	72.8±0.1	73.7±0.2	76.2±0.1	60.3±0.2	74.4±0.1	77.0±0.2	82.4±0.1
AA(%)	62.8±0.1	72.8±0.1	73.7±0.2	76.2±0.1	60.3±0.2	74.4±0.1	77.0±0.2	82.4±0.1
Kappa(%)	53.4±0.1	66.1±0.1	67.1±0.2	70.3±0.1	50.3±0.2	68.0±0.1	71.3±0.2	76.8±0.1

TABLE XIX
CLASSIFICATION RESULTS ON THE 2018 IEEE GRSS DFC DATASET

Index	RBF-SVM[9]	SS-CNN[26]	PPF-CNN[23]	3D-CNN[19]	HS-GAN[37]	3D-GAN[34]	AD-GAN[36]	SDC-GAN
OA(%)	55.3±0.2	69.8±0.1	71.1±0.2	74.4±0.1	62.1±0.3	73.4±0.1	76.9±0.1	82.8±0.2
AA(%)	26.6±0.4	48.5±0.3	49.4±0.4	52.7±0.2	35.1±0.7	58.4±0.3	64.5±0.1	73.1±0.2
Kappa(%)	40.6±0.5	60.7±0.3	62.8±0.2	66.9±0.1	48.6±0.4	66.0±0.1	70.7±0.1	78.2±0.1

J. Generalization Ability on Semantic Segmentation of RGB Remote Sensing Dataset

To verify the generalization ability on the semantic segmentation task of RGB remote sensing images, a medium-sized benchmark dataset of the RGB remote sensing image, Vaihingen dataset [52] is selected. Vaihingen is a public 2-D semantic labeling dataset from International Society for Photogrammetry and Remote Sensing. It contains 33 aerial images of size 2494 × 2064 with the ground sample distance of 9 cm, including five classes of impervious surfaces, building, low vegetation, tree, and car. Each image includes three spectral bands—near infrared, red, and green. We select 16 images for training and the remaining images for test. The quantitative results of SDC-GAN and the other seven comparison algorithms on the Vaihingen dataset are shown in Table XVIII.

As shown in Table XVIII, RBF-SVM and HS-GAN have lower classification results with all the OA, AA, and Kappa indexes below 70%. The OA results of PPF-CNN and 3D-CNN are 0.9% and 3.4% higher than that of SS-CNN, respectively. The OA results of 3D-GAN, AD-GAN, and SDC-GAN are 14.1%, 16.7%, and 22.1% higher than that of HS-GAN, respectively, and SDC-GAN has the highest classification results. The experimental results demonstrate the generalization ability and superiority of the proposed SDC-GAN method.

K. Classification Results on the 2018 IEEE GRSS DFC Dataset

We have added this more recent benchmark dataset, 2018 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest (DFC) dataset [53], to verify the effectiveness of the proposed method. The dataset was collected by an ITRES CASI 1500 sensor, covering a 380–1050-nm spectral range with 48 bands at the 1-m ground sample distance. The image size is 4172 × 1202. In the experiment, we use the top half of the image as the training set and the bottom half as the test set, where the classes of artificial turf and stadium seats are missing from the test set. The classification results of the remaining 18 classes are recorded. It is worth noting that the training set and test set are completely nonoverlapping. The classification results on this dataset are shown in Table XIX.

As shown in Table XIX, among the eight algorithms, the classification results of the other seven deep learning-based algorithms are higher than that of RBF-SVM. The OA results of PPF-CNN and 3D-CNN are 1.3% and 4.6% higher than that of SS-CNN, respectively. The OA results of 3D-GAN, AD-GAN, and SDC-GAN are 11.3%, 14.8%, and 20.7% higher than that of HS-GAN, respectively. Among these methods, SDC-GAN has the highest OA, AA, and Kappa results, which demonstrates the superiority of the proposed SDC-GAN method.

V. CONCLUSION

In this article, a novel SDC-GAN algorithm is proposed for HSI classification. The design of self-supervised network makes full use of abundant information in the unlabeled samples. The well-trained clusterer from the self-supervised network provides an excellent clustering representation for the sample division, which is beneficial for improving the generation ability of the generator and the classification performance of the discriminator in SDC-GAN. Specifically, in the generator, the data distribution of the specific branches is relatively balanced by dividing the samples of majority classes into multiple clusters, which alleviates the problem of mode collapse and improves the quality of generated samples. In the discriminator, the specific branches are customized for hard samples with similar characteristics; then, the classifiers added after these specific branches can further classify the hard samples and improve the classification accuracy. The experimental results on several HSI datasets demonstrate that SDC-GAN outperforms several state-of-the-art methods. In SDC-GAN, unlabeled samples are only utilized in the self-supervised pretext task. In the future, we will incorporate these unlabeled samples in the downstream classification task, so as to further improve the classification performance and generalization ability of SDC-GAN.

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for making the Vaihingen datasets publicly available, and also thank the IEEE GRSS Image Analysis and Data Fusion Technical Committee and the Hyperspectral Image Analysis Laboratory, University of Houston, Houston, TX, USA, for acquiring and providing the 2018 IEEE GRSS DFC dataset used in this article.

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